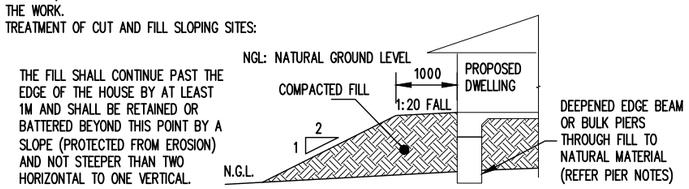
PROPOSED RESIDENTIAL DEVELOPMENT
AT LOT 3 (NO. 75) BEROWRA WATERS ROAD, BEROWRA, NSW

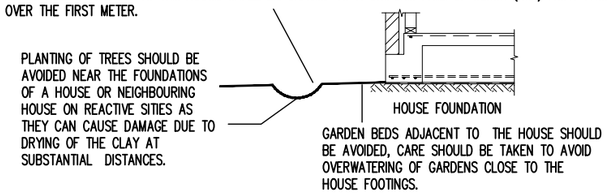
GENERAL

G1. THESE DRAWING SHALL BE READ IN CONJUNCTION WITH ALL ARCHITECTURAL AND OTHER CONSULTANTS DRAWING AND SPECIFICATIONS AND WITH SUCH OTHER WRITTEN INSTRUCTIONS AS MAY BE ISSUED DURING THE COURSE OF THE CONTRACT. ALL DISCREPANCIES SHALL BE REFEREED TO THE ENGINEER FOR DECISION BEFORE PROCEEDING WITH THE WORK.
G2. ALL DIMENSION SHOWN SHALL BE VERIFIED ON SITE. ENGINEERS DRAWINGS MUST NOT BE SCALED. (REFER ARCHITECTURAL DRAWINGS).
G3. THE STRUCTURAL ENGINEER SHALL PREPARE, SIGN AND ISSUE STRUCTURAL DRAWINGS FOR BUILDERS COMPULSORY IMPLEMENTATION DURING THE FULL OF THE CONSTRUCTION CONTRACT AND/OR ANY MAINTENANCE PERIOD CONTRACTS.
G4. DURING CONSTRUCTION THE CONTRACTOR SHALL BE RESPONSIBLE FOR MAINTAINING THE STRUCTURE IN A STABLE CONDITION AND ENSURE NO PART SHALL BE OVERSTRESSED UNDER CONSTRUCTION ACTIVITIES.
G5. ALL MATERIALS & WORKMANSHIP SHALL BE IN ACCORDANCE WITH CURRENT AUSTRALIAN STANDARDS AND SPECIFICATIONS.
G6. DURING THE CONSTRUCTION STAGE AND/OR ANY MAINTENANCE PERIOD, PROPER AND ADEQUATE CONTRACT WORKS INSURANCE AND PUBLIC LIABILITY INSURANCES IS COMPULSORY CLAIMS OR DAMAGE TO ANY ADJACENT PROPERTY OR BUILDING SHALL NOT BE RESPONSIBLE OF THE ENGINEER.
G7. THESE DRAWINGS ARE SIGNED SUBJECT TO A CERTIFICATE OF INSPECTION BEING ISSUED ISSUED BY ENGINEER. ALL PIER HOLES AND REINFORCEMENT SHALL BE INSPECTED BY THIS OFFICE PRIOR TO PLACING CONCRETE.
G8. SHOULD ANY STRUCTURAL ELEMENTS PRESENT DIFFICULTY WITH RESPECT TO CONTRACTIBILITY, SAFETY OR POTENTIAL HAZARDS, THE MATTERS SHALL BE REFERRED TO THE STRUCTURAL ENGINEER FOR RESOLUTION BEFORE PROCEEDING WITH THE WORK.
G9. TREATMENT OF CUT AND FILL SLOPING SITES:



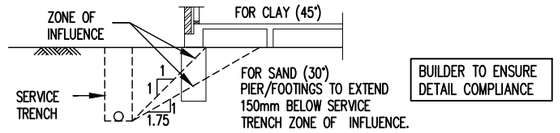
FOUNDATION AND FOOTING

F1. STRIP TOPSOIL CONTAINING ORGANIC MATTER. PROOF ROLL FILL SUB GRADE AND REMOVE ANY SOFT ZONES.
F2. WHERE ADDITIONAL FILL IS REQUIRED TO UNDERSIDE OF SLABS ON GROUND, NON COHESIVE MATERIALS SUCH AS SAND AND GRAVEL DUST SHALL BE PLACED BY "CONTROLLED" COMPACTION IN HORIZONTAL LAYERS OF 200mm (LOOSE) MAXIMUM DEPTH. THIS FILL SHALL BE COMPACTED TO AT LEAST 95% OF STANDARD MAXIMUM DRY DENSITY (SMDD) AND COMPACTED IN ACCORDANCE WITH AS 2870.
F3. FOR SLAB ON GROUND, SAND 50mm APPROXIMATE THICKNESS IS TO BE SPREAD AS A LEVELING LAYER AND WELL WATERED DOWN.
F4. DAMP-PROOFING MEMBRANE UNPUNCTURED AND TAPED AT LAPS, IS TO BE PLACED OVER THE SAND, SUFFICIENT MEMBRANE BEING PROVIDED AT EDGES TO RETURN UNDER BRICKWORK. WHERE NO BRICKWORK, TAPE MEMBRANE TO SIDE OF FOOTING BELOW GROUND.
F5. FOOTINGS ARE NOT TO BE CONSTRUCTED IN FILL MATERIAL, ALL STRUCTURAL FILL TO BE APPROVED BY GEOTECHNICAL ENGINEER.
F6. FOOTING HAVE BEEN DESIGNED FOR ALLOWABLE EQUAL UNIFORM BEARING CAPACITY OF 150KPa ON FIRM NATURAL GROUND. ALL FOUNDATION MATERIAL SHALL BE APPROVED FOR THIS BEARING CAPACITY BEFORE PLACING REINFORCEMENT.
F7. EXCAVATION SHALL CONTINUE UNTIL THE REQUIRED BEARING CAPACITY IS FOUND. THE OVER-EXCAVATION SHALL BE BACK-FILLED WITH A MASS CONCRETE MIX TO THE APPROVAL OF THE ENGINEER.
F8. REFER JOB SPECIFICATION DETAILS (PAGE 3) FOR SITE CLASSIFICATION. WE DISCLOSE THAT WE HAVE NOT VERIFIED THIS REPORT AND THAT WE RELY ON ITS FINDINGS. THIS OFFICE TAKE NO RESPONSIBILITY FOR VARIATIONS THAT MAY OCCUR DUE TO VARIATIONS IN SITE CONDITIONS.
F9. THE CONTRACTOR SHALL BE RESPONSIBLE FOR MAINTAINING EXCAVATIONS IN A STABLE CONDITION WITHOUT AFFECTING ADJACENT PROPERTIES OR SERVICES. WHERE REQUIRED, TEMPORARY SHORING SHALL BE PROVIDED TO THE SIDE OF FOOTING EXCAVATIONS.
F10. THE OWNERS ATTENTION SHOULD BE DRAWN TO APPENDIX B OF AS2870 "PERFORMANCE REQUIREMENTS AND FOUNDATION MAINTENANCE" ON COMPLETION OF THE JOB.
F11. THE SITE SHALL BE GRADED OR DRAINED SO THAT WATER CANNOT POND AGAINST OR NEAR THE HOUSE. THE GROUND IMMEDIATELY ADJACENT TO THE HOUSE SHALL BE GRADED TO A UNIFORM FALL OF 50mm (min) AWAY FROM THE HOUSE OVER THE FIRST METER.



FOUNDATION PIERS

P1. THE CONTRACTOR SHALL INVESTIGATE THE PRESENCE OF ANY EXISTING SERVICES IN THE GROUND LIKELY TO BE AFFECTED BY THE PILING OPERATIONS.
P2. PILES AND PILING ARE IN ACCORDANCE WITH AS 2150 AND ANY OTHER RELEVANT CODES.
P3. DURING THE DRILLING THROUGH EXISTING FILL, OBSTRUCTION MAY BE EXPECTED.
P4. IF GROUNDWATER IS ENCOUNTED DURING THE PILING, THE ENGINEER TO BE CONTACTED IMMEDIATELY.
P4. ENSURE THAT MINIMUM PIER DEPTH IS 600mm BELOW THE FOOTING TRENCH, U.N.O.
P5. ALL PIERS TO BE POURED SEPARATE FROM FOOTINGS AND SLABS.
P6. ALL PIERS ARE TO BE INSPECTED TO ENSURE THEY ARE CLEANED AND FREE OF LOOSE MATERIAL AND WATER PRIOR TO POURING CONCRETE, WHICH SHOULD BE WITH MINIMAL DELAY AND ON THE SAME DAY AS BORING.
P7. PIER TOPS ARE TO BE FINISHED SMOOTH. ITS IS RESPONSIBILITY OF CONTRACTOR TO ENSURE PIER TOPES ARE CLEAN AND FREE OF FOREIGN MATERIAL PRIOR TO THE LAYING OF THE MEMBRANE AND CONCRETE SLAB. ENGINEER SPOT CHECK DOES NOT RELEASE THE CONTRACTOR FROM THIS RESPONSIBILITY.
P8. ALL EDGE BEAMS & INTERNAL BEAMS TO BE FOUNDED ON CONSISTENT NATURAL STRATA OF (MIN.) BEARING CAPACITY 150KPa OTHERWISE, ø400 PIERS AT 1800 CTS (MAX.) ARE REQUIRED TO BY-PASS FILL & ACHIEVE UNIFORM BEARING CAPACITY (MIN. 250kPa) ON NATURAL STRATA. REFER STRUCTURAL PLANS PIER DIAMETER AND LOCATIONS.
P9. IF ANY PIER (OR FOOTING) ENCOUNTERS ROCK OR SHALE, ALL EDGE BEAMS ARE TO BE PIERED TO ROCK OR SHALE- ø300 PIERS @ 2.2m CTS.
P10. WHERE DEPTH OF PIERS EXCEED 2.5m, ENGINEER TO BE NOTIFIED FOR DECISION.
P11. AT THE COMPLETION OF PIERS, THE CONTRACTOR SHALL PROVIDE AN AS-BUILT SURVEY AND CERTIFICATION THAT ALL PIERS ARE WITHIN SPECIFIED TOLERANCES.



SERVICE TRENCH DETAILS

REINFORCED CONCRETE

C1. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS 3600 CURRENT EDITION, EXCEPT WHERE VARIED BY THE CONTACT DOCUMENTS.
C2. CONCRETE QUALITY (ENGINEER TO APPROVE ANY ADMIXTURE USED IN CONCRETE MIX)

ELEMENT	SLUMP	MAX SIZE AGGREGATE	Fc (28 DAYS)
SLAB	80mm	20mm	32 MPa
FOOTINGS	80mm	20mm	32 MPa
PIERS	100mm	20mm	32 MPa

C3. ALL CONCRETE PLACED SHALL BE WELL VIBRATED AND COMPACTED USING AN APPROVED VIBRATOR AND TECHNIQUE.
C4. CONSTRUCTION JOINTS WHERE NOT SHOWN SHALL BE LOCATED TO THE APPROVAL OF THE ENGINEER.
C5. SIZES OF CONCRETE ELEMENTS DO NOT INCLUDE THICKNESS OF ANY APPLIED FINISHES.
C6. NO HOLES OR CHASES OTHER THAN THOSE SHOWN ON THE STRUCTURAL DRAWINGS SHALL BE MADE IN CONCRETE MEMBERS. WITHOUT THE PRIOR APPROVAL OF THE ENGINEER.
C7. WATER IS NOT TO BE ADDED TO THE CONCRETE ON SITE SO AS TO INCREASE THE SLUMP ABOVE THAT SPECIFIED.
C8. THE CONTRACTOR SHALL NOTIFY THE ENGINEER 24 HOURS BEFORE REINFORCEMENT IS COMPLETED. THE CONTRACTOR SHALL ALLOW AFTER THE COMPLETION OF THE REINFORCEMENT TWO HOURS FOR THE ENGINEERS INSPECTIONS, UPON APPROVAL CONCRETE MAY BE ORDERED.
C9. CONCRETE TO BE CURED A MINIMUM OF 7 DAYS AFTER POURING.
C10. CURING OF ALL CONCRETE SHALL BE ACHIEVED BY KEEPING SURFACES COMPLETELY WET FOR A PERIOD OF 3 DAYS, AND PREVENTION OF LOSS OF MOISTURE FOR A TOTAL OF 7 DAYS FOLLOWED BY A GRADUAL DRYING OUT. APPROVED SPRAYED ON CURING COMPOUNDS THAT COMPLY WITH AS 3799 MAY BE USED WHERE FLOOR FINISHES WILL NOT BE AFFECTED (REFER MANUFACTURERS SPECIFICATION). POLYTHENE SHEETING OR WET HESSIAN OR SIMILAR PRODUCT MAY BE USED TO RETAIN CONCRETE MOISTURE WHERE PROTECTED FROM WIND AND TRAFFIC.
C11. FORMWORK TO BEAMS AND SLABS SPANNING GREATER THAN 5m SHALL BE PRE-CAMMBERED UPWARDS BY 1/500 OF CLEAR SPAN IN EACH SPAN UNLESS NOTED DIFFERENTLY ON THE DRAWINGS.
C12. SHRINKAGE CRACKING IS ALMOST INEVITABLE & DOES NOT REPRESENT FAILURE. HOWEVER, IS OF CONCERN UNDER BRITTLE FLOOR COVERINGS DAMAGE MAY BE REDUCED BY USING FLEXIBLE MORTARS & GLUES FOR FIXING TILES & FIXING OPERATION MUST BE DELAYED AS LATE AS POSSIBLE.
C13. CONSTRUCTION SUPPORT PROPPING IS TO BE LEFT IN PLACE WHERE NEEDED O AVOID OVERSTRESSING THE STRUCTURE DUE TO CONSTRUCTION LOADING. NO BRICKWORK OR PARTITION WALLS ARE TO BE CONSTRUCTED ON SUSPENDED LEVELS UNTIL ALL PROPPING IS REMOVED AND THE SLAB HAS ABSORBED ITS DEAD LOAD DEFLECTION.

REINFORCEMENT

R1. MINIMUM CLEAR CONCRETE COVER REINFOREMENT INCLUDING TIES AND STIRRUPS (OTHER THAN RESIDENTIAL SLABS ON GROUND OR FOOTINGS) SHALL BE FOLLOWS UNO.

EXPOSURE CLASSIFICATION	MINIMUM COVER (mm) CONCRETE STRENGTH (Fc)				
	20 MPa	25 MPa	32 MPa	40 MPa	> 50 MPa
A1	20	20	20	20	20
A2	(50)	30	25	20	20
B1	-	(60)	40	30	25
B2	-	-	(65)	45	35
C	-	-	-	(70)	50

FOR BRACKETED FIGURES REFER TO AS3600 CURRENT EDITION TABLE 4.10.3.2

R2. RESIDENTIAL SLAB ON GROUND AND FOOTINGS COVER REQUIREMENTS:
(MINIMUM CONCRETE GRADE N20)
- UNPROTECTED GROUND: 40m
- EXTERNAL EXPOSURE: 40mm
- MEMBRANE IN CONTACT WITH GROUND: 30mm
- INTERNAL SURFACE: 20mm
- STRIP AND PAD FOOTING: 40mm

R3. REINFORCEMENT SYMBOLS:
N - DENOTES DEFORMED GRADE 500 NORMAL DUCTILITY REINFORCING BARS TO AS/NZS4671.
R - DENOTES PLAIN ROUND GRADE 250 NORMAL DUCTILITY REINFORCING BARS TO AS/NZS4671.
SL - DENOTES DEFORMED GRADE 500 LOW DUCTILITY REINFORCING MESH TO AS/NZS4671.
RL - DENOTES DEFORMED GRADE 500 LOW DUCTILITY REINFORCING MESH TO AS/NZS4671.
L-TM - DENOTES DEFORMED GRADE 500 LOW DUCTILITY TRENCH MESH TO AS/NZS4671.
R4. REINFORCEMENTS IS REPRESENTED DIAGRAMMATICALLY, IT IS NOT NECESSARILY SHOWN IN TRUE PROJECTION.
R5. SPLICES IN REINFORCEMENTS SHALL BE MADE ONLY IN POSITIONS SHOWN OTHERWISE APPROVED BY ENGINEER.
R6. FABRIC REINFORCEMENT SHALL HAVE SPLICES MADE SO THAT THE OVERLAP, MEASURED BETWEEN THE OUTERMOST TRANSVERSE WIRES OF EACH SHEET OR FABRIC IS NOT LESS THAN THE SPACING OF THOSE WIRES PLUS 25mm.
R7. AROUND THE ENTIRE EDGE OF THE SLAB, THERE MUST BE AN INTACT EDGE WIRE OF THE FABRIC PARALLEL TO AND 20mm FROM THE FORM BOARD.- PRESENT, PROVIDE AN ADDITIONAL 1-N12 TO EDGE.
R8. WELDING OF REINFORCEMENT SHALL NOT BE PERMITTED UNLESS SHOWN ON STRUCTURAL DRAWINGS OR APPROVED BY ENGINEER.
R9. TOP AND BOTTOM REINFORCEMENT IN SLABS SHALL BE SUPPORTED ON APPROVED PLASTIC TIPPED CHAIRS, IN BOTH DIRECTIONS AT MAXIMUM CENTRES OF 600mm FOR 10mm DIA BARS, 900mm FOR 12mm AND 16mm DIA BARS, 1200mm FOR 20mm DIA BARS AND 750mm CENTRES FOR FABRIC.
R10. TRUSSED ROOF NOTE: SLAB HAS BEEN DESIGNED FOR ROOF LOADING TO BE SUPPORTED BY PROPRIETARY TRUSSES ONTO EXTERNAL WALLS ONLY.
R11. ARTICULATION NOTE: THIS SLAB IS DESIGNED FOR ARTICULATED MASONRY VENEER TYPE CONSTRUCTION AND ARTICULATION JOINTS ARE TO BE PROVIDED AS PER THE B.C.A. VOL.2 CURRENT EDITION
R12. DEEPEND BEAM NOTE: THE FULL EXTENT OF DEEPENED EDGE BEAMS AND ANY SET-DOWNS ARE TO BE CARRIED OUT IN ACCORDANCE WITH THE ARCHITECTURAL DETAILS
R13. HWS AND ACU NOTE: PADS ARE TO BE LOCATED AS PER ARCHITECTURAL DRAWINGS. WHEN IN FILL, PLACE MASS CONCRETE REINFORCEMENT COVER TABLE PIER UNDER CONCRETE PADS.
R14. CONDUITS, PIPES, ECT. SHALL ONLY BE PLACED IN THE MIDDLE ONE THIRD OF SLAB DEPTH AND SPREAD AT NOT LESS THAN 3 DIAMETERS.
R15. ALL REENTRANT CORNERS AT PENETRATIONS FOR SLUMP, PITS, COLUMN BLOCKOUTS AND THE LIKE, TO HAVE 2N12 TRIMER BARS PLACED AT 45 DEGREE TO CORNER OR IN EACH DIRECTION AT CORNERS U.N.O. TRIMERS BARS TO BE TIED UNDER SIDE OF TOP MESH.

REINFORCEMENT LAP TABLE						
BAR SIZE	N12	N16	N20	N24	N28	N32
LAP LENGTH	450	600	800	1200	1350	1650

A3 0 1 2 3 4 5 6 7 8 9 10

ISSUE	DESCRIPTION	DATE	ISSUE	DESCRIPTION	DATE
A	ISSUED FOR CC	06.10.26			

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PROPOSED RESIDENCE
LOT 3 (NO. 75) BEROWRA WATERS ROAD
BEROWRA, NSW
FOR - VIKAS KUMAR RANGI
& POONAM CHAUDHARY

BUILDER

CONSTRUCTION NOTES - 1

SCALE: 1:100, 1:20 U.N.O.

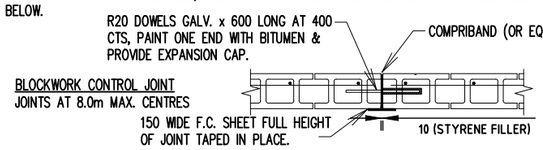
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BRICKWORK AND BLOCKWORK

B1. ALL WORKMANSHIP AND MATERIALS TO BE IN ACCORDANCE WITH THE CURRENT AS 3700 AND SAA MASONARY CODES.
B2. INTERNAL BRICKWORK BUILT OFF THE SLAB SHALL BE LAID ON TWO LAYERS OF 'ALCOR' OR '3 PLY MALTHOID' OR SIMILAR SLIP JOINT MATERIAL.
B3. LOAD BEARING MASONRY SHALL COMPLY WITH AS 3700
B4. MASONRY AND MORTAR DURABILITY SHALL COMPLY WITH THE B.C.A CLASS 1 AND 10 BUILDINGS VOLUME 2.
B5. MASONRY SHALL BE ARTICULATED IN ACCORDANCE WITH THE B.C.A CLASS 1 AND 10 BUILDINGS VOLUME 2.
B6. MASONRY WALLS MUST NOT BE BUILT ON CONCRETE SLABS OR BEAMS UNTIL ALL FORMWORK/PROPS SUPPORTING THESE SLABS AND BEAMS HAVE BEEN REMOVED.
B7. ALL BLOCKS SHALL HAVE A MINIMUM COMPRESSIVE STRENGTH OF 25 MPa.
B8. CHASES AND HOLES SHALL NOT BE MADE IN BLOCKWORK WITHOUT THE PRIOR APPROVAL OF SUPERINTENDENT.
B9. PROVIDE CLEAN OUT HOLES AT THE BOTTOM OF ALL CORES FILLED. ALL MORTAR SHALL BE REMOVED FROM BOTTOM OF WALL BEFORE CLEAN OUT HOLES ARE FILLED FOR GROUTING.
B10. ALL CORES SHALL BE CLEANED AND VERTICAL BARS TIED TO STARTER BARS.
B11. CONCRETE TO WALLS TO BE THOROUGHLY VIBRATED AND POURED IN LIFTS OF 1800 MAXIMUM. SECOND LIFT TO BE VIBRATED INTO THE FIRST LIFT.
B12. THE TOP COURSE IN HOLLOW BLOCK WALLS SHALL BE LAID IN SOLID BLOCK.
B13. PROVIDE CLEAN-OUT HOLES AT BASE OF ALL WALLS AND ROD CORE HOLES TO REMOVE PROTRUDING MORTAR FINS.
B14. CORE FILLING GROUT SHALL HAVE MINIMUM CHARACTERISTIC STRENGTH OF 25 MPa, 10mm AGGREGATE, 230mm SLUMP +/- 30mm. THOROUGHLY COMPACT CORE FILL GROUT WITH A MECHANICAL VIBRATOR.
B15. PROVIDE 55mm COVER TO REINFORCING BARS FROM THE OUTSIDE FACE OF THE BLOCKWORK.
B16. BACKFILL TO RETAINING WALLS IS TO BE FREE DRAINING GRANULAR MATERIAL UNLESS NOTED OTHERWISE. PROVIDE A SUBSOIL DRAIN TO WEEP HOLES.
B17. ALL MASONRY WALLS AND PIERS SUPPORTING SLABS AND BEAMS SHALL HAVE A PRE-GREASED GALVANISED STEEL SLIP JOINT BETWEEN THE TOP OF THE MASONRY AND THE CONCRETE SLAB SOFFIT UNLESS NOTED OTHERWISE.
B18. PROVIDE VERTICAL JOINTS AT 8m MAX. CENTERS, AND 5m MAXIMUM DISTANCE FROM CORNERS IN ALL MASONRY WALLS. REFER DETAIL BELOW.



DURABILITY CLASSIFICATION			
CATEGORY (DURABILITY)	WALL TIES AS3700	SALT ATTACK RESISTANCE OF BRICK	MORTAR AS3700
LESS THAN 1km FROM BREAKING SURF OR LESS THAN 100m FROM SALT WATER NOT SUBJECT TO BREAKING SURF	R4 (STAINLESS OR POLYMER)	EXPOSURE – AS3700	M4 (1:4)
1km OR LESS THAN 10km FROM BREAKING SURF OR 100m OR LESS THAN 1km FROM SALT WATER NOT SUBJECT TO BREAKING SURF	R3	EXPOSURE (GENERAL PURPOSE – AS3700) ✳	M3 (1:5)
ALL OTHER AREAS	R2	GENERAL PURPOSE – AS3700	M2 (1:2:8)
THE TABLE ABOVE IS A PART SUMMARY OF THE REQUIREMENTS OF CURRENT EDITION OF AS 3700 & AS 4100. IT IS RECOMMENDED TO REFER THESE AUSTRALIAN STANDARDS FOR MORE OPTIONS/INFORMATIONS. ✳ REFER NOTES: MANDATORY REQUIREMENTS FOR SALINITY AFFECTS.			

DRAINAGE REQUIREMENTS

D1. ALL WORKMANSHIP AND MATERIAL SHALL BE IN ACCORDANCE WITH CURRENT EDITIONS OF AS2870, AS/NZS 2032 INTALL OF PVC PIPES AND AS/NZS 3500 PLUMBING & DRAINAGE.
D2. PLUMBING TRENCHES SHALL BE SLOPED AWAY FROM THE HOUSE AND SHALL BE BACKFILLED WITH CLAY IN THE OP 300mm WITHIN 1.5m OF THE HOUSE. THE CLAY USED FOR BACKFILLING SHALL BE COMPACTED. WHERE PIPES PASS UNDER THE FOOTING SYSTEM, THE TRENCH SHALL BE BACKFILLED WITH CLAY OR CONCRETE TO RESTRICT THE INGRESS OF WATER BENEATH THE FOOTING SYSTEM.
D3. DRAINAGE SHALL BE CONSTRUCTED TO AVOID WATER PONDING AGAINST OR NEAR THE FOOTING.
D4. ECAVATION NEAR THE EDGE OF THE FOOTING SYSTEM SHALL BE BACKFILLED IN SUCH A WAY AS TO PREVENT ACESS OF WATER TO THE FOUNDATION.
D5. WATER RUN-OFF SHALL BE COLLECTED AND CHANNELLED AWAY FROM THE HOUSE DURING CONSTRUCTION.
D6. PENETRATIONS OF THE EDGE BEMS AND FOOTING BEAMS ARE TO BE AVOIDED, BUT WHERE NECESSARY SHALL BE SLEEVED TO ALLOW FOR MOVEMENT.
D7. CONNECTION OF STORMWATER DRAINS AND WASTE DRAINS SHALL BE INCLUDED FLEXIBLE CONNECTIONS.

STRUCTURAL STEELWORK

S1. ALL MATERIAL, WORKMANSHIP, FABRICATION AND ERECTION SHALL BE IN ACCORDANCE WITH AS 4100, AS 1538, AS 1554 AND RELEVANT ASSOCIATED STANDARDS.
S2. UNLESS NOTED OTHERWISE, ALL STEEL SHALL BE IN ACCORDANCE WITH AS1204 GRADE 300. ALL STEEL HOLLOW SECTIONS SHALL BE IN ACCORDANCE WITH AS1163 GRADE 350. ALL PRESSED METAL PURLINS AND GIRTS SHALL BE IN ACCORDANCE WITH AS1538 GRADE 450.
S3. THREE SETS OF STEELWORK SHOP DETAIL DRAWINGS SHALL BE SUBMITTED TO THE ENGINEER FOR PPROVAL PRIOR TO COMMENCEMENT OF ANY FABRICATION. THE CHECK SHALL NOT COVER LAYOUT AND MEMBER DIMENSION.
S4. UNLESS NOTED OTHERWISE, USE:
A) 10mm GUSSET, BASE AND END PLATES.
B) M20 8.8/S BOLTS. HIGH STRENGTH STRUCTURAL (AS1252)
C) 6mm CONTINUOUS FILLET WELDS MADE WITH E41XX MILD ELECTRODES.
D) ALL BUT WELDS SHALL BE COMPLETE PENETRATION BUTT WELD TO A1554.1
S5. SITE WELDING OF HOT DIP GALVANIZED STEEL IS PERMISSIBLE IF UPON COMPLETION THE WELDS ARE TREATED WITH THE APPROPRIATE COATING FOR SEVERE AS PER THE B.C.A AND AS/NZS 2312.
S6. BEAMS SUPPORTED ON BRICKWORK (BEARING NOTED ON PLAN) TO HAVE INCOMPRESSIBLE PACKING AS REQUIRED UNDER THE ENDS OF THE BEAM TO ENSURE EVEN BEARING ON THE BRICKWORK.
S7. THE BUILDER SHALL PROVIDE ALL CLEATS AND DRILL ALL HOLES NECESSARY FOR FIXING STEEL TO STEEL AND TIMBER AND OTHER ELEMENTS TO STEEL WHETHER OR NOT DETAILED IN THE DRAWINGS.
S8. WELDS TO BE PROPERLY CHIPPED TO REMOVE SLAG.
S9. STEELWORK NOT ENCASED SHALL HAVE THE FOLLOWING SURFACE TREATMENTS

EXPOSURE CLASSIFICATION	STEELWORK PROTECTION REQUIRED
A1/A2	POWDER TOOL CLEAN TO AS1627 CLASS 1, 1 COT ALKYD PRIMER (ZINC PHOSPHATE)
B1	ABRASIVE BLAST TO AS1627 CLASS 2.5, 1 COT ZINC PHOSPHATE SILICATE
B2	HOT DIPPED GALVANISED TO AS4680, 600gm/M2

S11. THE BUILDER SHALL PROVIDE ALL CLEATS AND DRILL ALL HOLES NECESSARY FOR FIXING STEEL TO STEEL AND TIMBER AND OTHER ELEMENTS TO STEEL WHETHER OR NOT DETAILED IN THE DRAWINGS.
S12. ALL STEELWORK SHALL HAVE ADEQUATE TEMPORARY BRACING PROVIDED TO STABILIZE THE STRUCTURE DURING CONSTRUCTION.
S13. THE FABRICATION AND ERECTION OF THE STRUCTURAL STEEL WORK SHALL BE SUPERVISED BY QUALIFIED ENGINEER, EXPERIENCE IN SUCH SUPERVISION, TO ENSURE THAT ALL REQUIREMENTS OF THE OF THE DESIGN ARE MET.
S14. SURFACE FINISHES FOR ALL STRUCTURAL STEELWORK TO BE IN ACCORDANCE WITH THE ARCHITECTURAL SPECIFICATION.
S15. THE INSTALLATION OF GALINTELS AND 'I' BARS TO BE STRICTLY IN ACCORDANCE WITH MANUFACTURERS SPECIFICATIONS. PLACE GALINTEL 'I' BAR OVER OPENING ALLOWING A MINIMUM OF 200mm END BEARING EACH END.GALINTEL 'I' BAR MUST BE PROPPED BEFORE BRICKLAYING. FROM 2.4m TO 3.3m SPAN 2 PROPS. FROM 3.6m TO 4.5m SPAN 3 PROPS. FROM 4.8m TO 5.7m SPAN 4 PROPS. SPACED EQUALLY ALONG THE LENGTH AND UNDER THE BASE OF THE BAR. WHEN LAYING BRICKS MORTAR MUST BE APPLIED TO ALL BRICK FACES COMING IN CONTACT WITH THE 'I' BAR. PROPS TO REMAIN IN PLACE UNTIL MORTAR ACHIEVES FULL STRENGTH (7 DAYS MIN.). A MINIMUM 1:4 MORTAR MIX IS TO BE USED AND APPLIED TO ALL FACES BETWEEN STEEL AND BRICKS (VERTICAL AND HORIZONTAL LEGS) AND BETWEEN BRICKS ABOVE THE STEEL SECTION.

MANDATORY REQUIREMENTS FOR SALINITY AFFECTS

FOR HOUSE SLAB FOTING AND BRICKWORKS
SN1. FOR SLAB ON GROUND CONSTRUCTION, A LAYER OF SAND AT LEAST 50mm DEEP UNDER THE SLAB MUST BE PROVIDED.
SN2. A HIGH IMPACT DAMP PROOF MEMBRANE (RATHER THAN A VAPOUR PROOF MEMBRANE) MUST BE LAID UNDER SLAB (NSW BCA 3.2.2.6)
SN3. THE DAMP PROOF MEMBRANE MUT EXTEND TO THE OUTSIDE FACE OF THE EXTERNAL EDGE BEAM UP TO THE FINISHED GROUND LEVEL.
SN4. ALL CLASS 32MPA (N32) CONCRETE MUST BE USED OR A SULPHATE RESISTING. TYPE SR CEMENT WITH A WATER CEMENT RATIO OF 0.5 MUST BE USED.
SN5. IT IS IMPERATIVE THAT WATER IS NOT ADDED AT THE CONSTRUCTION SITE FOR 32 MPA CONCRETE AS THESE WILL REDUCE THE CONCRETE STRENGTH.
SN6. SLABS MUST BE VIBRATED AND CURED FOR A MINIMUM OF THREE DAYS. CARE MUST BE TAKEN NOT TO OVER VIBRATE THE CONCRETE DURING PLACEMENT, AS SEGREGATION OF THE CONCRETE AGGREGATES WILL OCCUR.
SN7. THE MINIMUM COVER TO REINFORCEMENT MUST BE 50MM FROM UNPROTECTED GROUND. CHAIRS INCLUDING LATERAL SUPPORTS SHOULD BE IN POSITION PRIOR TO INSPECTION AND SUBSEQUENTLY DURING POURING OF THE CONCRETE.
SN8. THE MINIMUM COVER TO REINFORCEMENT MUST 30MM FROM THE MEMBRANE IN CONTACT WITH THE GROUND.
SN9. THE MINIMUM COVER TO REINFORCEMENT MUST BE 50MM FOR STRIP FOOTINGS AND BEAMS IRRESPECTIVE OF WHETHER A DAMP PROOF MEMBRANE IS USED.
SN10. APPROVED ADMIXTURES FOR WATERPROOFING AND/ OR CORROSION PREVENTION MAY BE USED.
SN11. THE FOLLOWING MEASURES MUST BE USED FOR BRICKWORK:

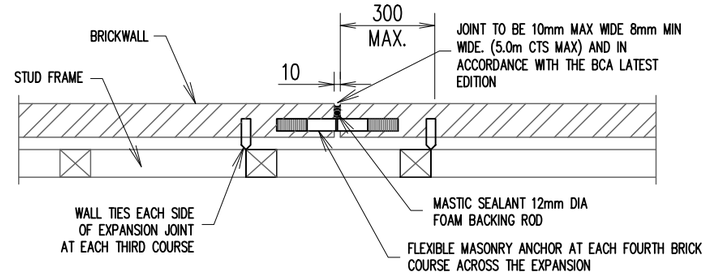
- THE DAMP PROOF COURSE MUST CONSIST OF POLYETHYLENE OR POLYETHYLENE COATED METAL AND BE CORRECTLY PLACED. (NSW BCA 3.3.4.4).
- EXPOSURE CLASS MASONRY UNITS MUST BE USED BELOW THE DAMP PROOF COURSE LEVEL. (CLAUSE 3.3.1.5(B) AND TABLE 3.3.1.1 OF THE BCA). APPROPRIATE MORTAR AND MIXING RATIO MUST BE USED WITH EXPOSURE CLASS MASONRY UNITS. (CLAUSE 3.3.1.6 OF THE BCA).
- APPROVED ADMIXTURES FOR WATERPROOFING AND/ OR CORROSION PREVENTION MAY BE USED. WORKMANSHIP AND MATERIAL SHALL BE IN ACCORDANCE WITH CURRENT EDITIONS OF AS2870, AS/NZS 2032 INTALL OF PVC PIPES AND AS/NZS 3500 PLUMBING & DRAINAGE.

TIMBER NOTES

T1. ALL TIMBER CONSTRUCTION SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF AS 1684 AND AS 1720.
T2. ALL TIMBER USED HAVE BEEN STRESS GRADED BY VISUAL OR MECHANICAL MEANS IN ACCORDANCE WITH APPROPRIATE AUSTRALIAN STANDARD.
T3. ALL SOFTWOOD MINIMUM GRADE F7 AND HARDWOOD MINIMUM GRADE F14 U.N.O.
T4. EXTERNAL TIMBER TO BE EITHER HARDWOOD DURABILITY CLASS I OR II OR IMPREGNATED GRADE F7. PRESSURE TREATED TO AS1684 AND RE-DRILLED PRIOR TO USE. SUPPLEMENTARY TREATMENT SHALL BE APPLIED TO ALL CUT SURFACES.
T5. ALL BOLTS SHALL BE 12mm DIAMETER STEEL UNLESS NOTE OTHERWISE. BOLT HOLES TO BE DRILLED EXACT SIZE. WASHERS UNDER HEADS & NUTS TO BE AT LEAST 2.5 TIMES BOLT DIAMETER.
T6. ALL TIMBER JOINTS AND NOTCHES TO BE 100mm MINIMUM FROM LOOSE KNOTS. SEVERE SLOPING GRAIN, GUM VEINS OR OTHER MINOR DEFECTS.
T7. ALL DESIGN, WORKMANSHIP & MATERIALS SHALL BE IN ACCORDANCE WITH NATIONAL TIMBER FRAMING CODE AS1684 CURRENT EDITION WITH AMENDMENTS, EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.
T8. TIMBER SIZES, CONNECTIONS AND BRACING WALL SHALL BE TO FRAME MANUFACTURER'S DETAILS & SPECIFICATIONS & SHALL BE IN ACCORDANCE WITH AS1684. TIMBER FRAMING OUTSIDE THE SCOPE OF AS1684 SHALL BE REFEREED TO THE SUPERINTENDENT FOR A DECISION BEFORE PROCEEDING.
T9. ALL WALLS, ROOF BRACING & ANCHOR DETAILS WHERE NOT SHOWN ON DRAWINGS SHALL BE IN ACCORDANCE WITH AS1684 & BCA REQUIREMENTS.
T10. THREE COPIES OF SHOP DRAWINGS ARE TO BE SUBMITTED TO THE ENGINEER FOR APPROVAL CLEARLY SHOWING THE DESIGN LOADS ON THE ROOF AND CEILING AND TRUSS NODE POINT LOADS AND PRECAMBER. DRAWINGS SHALL BE SUBMITTED MINIMUM 14 DAYS PRIOR TO COMMENCEMENT OF ANY FABRICATION.
T11. TIMBER TREATED IN ACCORDANCE WITH A.S. 1604 SHALL HAVE FOLLOWING HAZARD LEVEL. INTERIOR ABOVE GROUND (H2), EXTERIOR ABOVE GROUND (H3) AND EXTERIOR IN GROUND (H4 OR H5).

WIND/WALL TIE CLASSIFICATION				
WIND		WALL TIES	HORIZONTAL SPACING (MM)	VERTICAL SPACING (MM)
CLASS	(Vp)			
N1	W28N1	LIGHT DUTY	600	600
N2	W33N2	MEDIUM DUTY	600	600
N3	W41N3	MEDIUM DUTY	600	430 (5 COURSE)
WALL TIE SPACINGS AROUND OPENINGS 300 c/c EACH WAY				
POLYMER WALL TIES RATED "LIGHT DUTY ONLY" (W28N1)				
(Vp = PERMISSABLE STRESS METHOD)				

ARTICULATION JOINT DETAIL FOR BRICK VENEER



FOR CAVITY BRICK WALL AND SINGLE LEAF OR DOUBLE MASONRY ARTICULATION JOINTS DETAILS IN ACCORDANCE WITH THE BCA LATEST EDITION.

IMPORTANT NOTE:
"EFFECTS OF TREES + 'ABNORMAL' FOUNDATION MOISTURE CONDITIONS"

THIS SLAB LAYOUT DESIGN DOES NOT INCLUDE THE EFFECTS OF PREVIOUS, EXISTING OR FUTURE TREES, THEREFORE, IF ANY TREES WERE REMOVED OR ARE PRESENT WITHIN THE VICINITY OF THIS PROPOSED RESIDENCE OR ARE CONSIDERED FOR FUTURE PLANTING, THE DESIGN ENGINEER MUST BE ADVISED FOR RE-CONSIDERATION OF THIS DESIGN. ALSO, THIS SLAB LAYOUT DESIGN DOES NOT ACCOMMODATE GROUND MOVEMENTS GENERATED BY "ABNORMAL" FOUNDATION MOISTURE CONDITIONS. FOR A BETTER UNDERSTANDING OF THESE REQUIREMENTS, THE BUILDERS/OWNERS ATTENTION IS DRAWN TO:

1. APPENDIX CH OF AS2870-2011 "GUIDE TO DESIGN OF FOOTINGS FOR TREES.
2. APPENDIX B OF AS2870-2011 "PERFORMANCE REQUIREMENTS AND FOUNDATION MAINTENANCE"
3. CSIRO PAMPHLET "GUIDE TO HOME OWNERS ON FOUNDATION MAINTENANCE AND FOOTING PERFORMANCE".

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PROPOSED RESIDENCE
LOT 3 (NO. 75) BEROWRA WATERS ROAD
BEROWRA, NSW
FOR – VIKAS KUMAR RANGI
& POONAM CHAUDHARY

BUILDER

CONSTRUCTION NOTES – 2

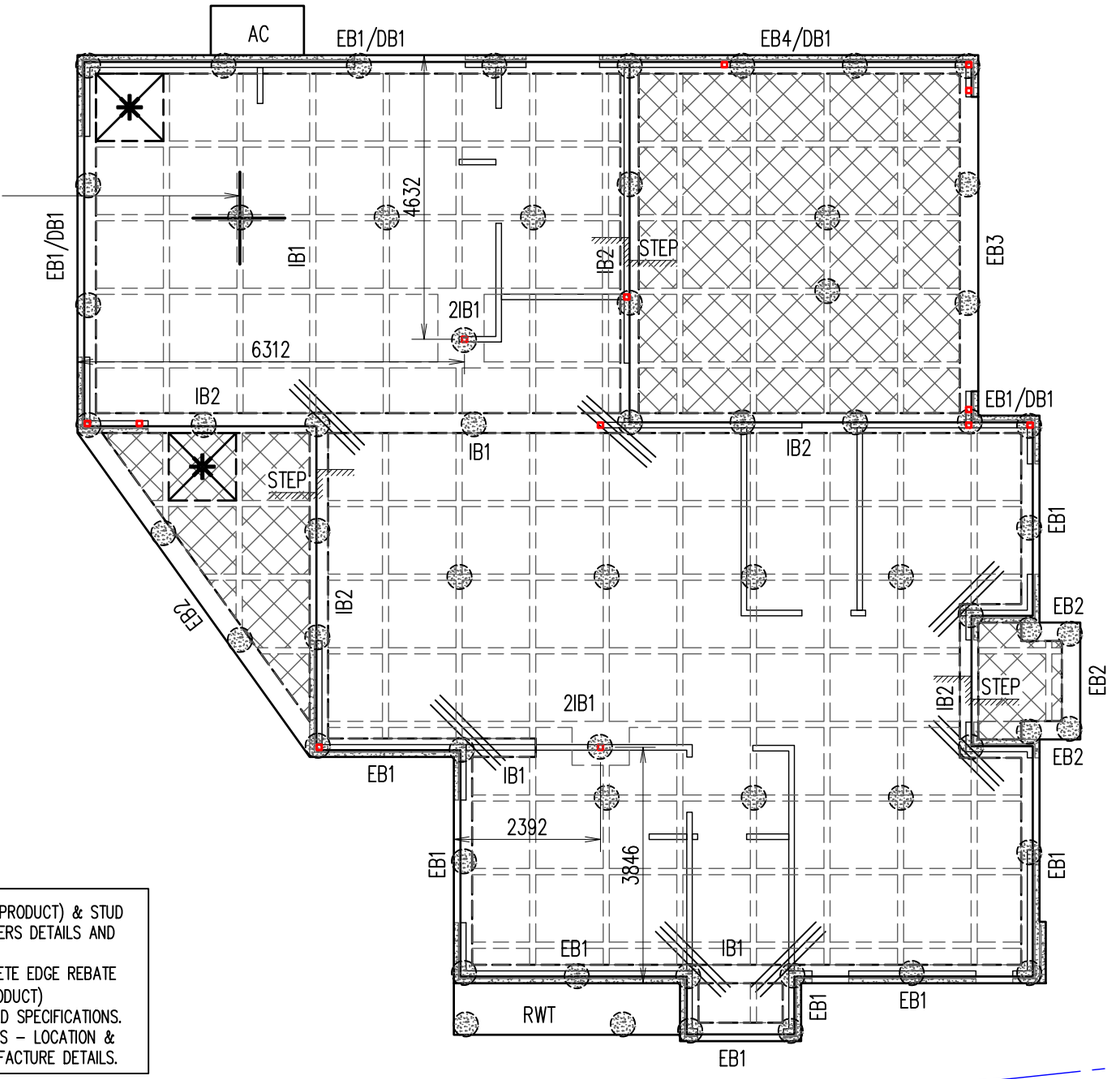
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PROVIDE ADDITIONAL 1N12 (2000 LONG) AT ALL INTERNAL PIER LOCATION – BOTH WAYS FIXED TO UNDERSIDE OF THE TOP FABRIC MESH. NOT SHOWN ON OTHERS INTERNAL PIERS FOR CLARITY. TYPICAL



1. INSTALL HEBEL (OR SIMILAR PRODUCT) & STUD WALLS AS PER MANUFACTURERS DETAILS AND SPECIFICATIONS.
2. BUILDER TO CONFIRM CONCRETE EDGE REBATE WITH HEBEL (OR SIMILAR PRODUCT) MANUFACTURERS DETAILS AND SPECIFICATIONS.
3. PROVIDE ARTICULATION JOINTS – LOCATION & DETAIL AS PER HEBEL MANUFACTURE DETAILS.

JOB SPECIFICATION	
SITE CLASSIFICATION	**P (DESIGN H1)**
TYPE OF CONSTRUCTION	**ARTICULATED HEBEL VENEER**
SALINITY AFFECTED	**YES – REFER PAGE 2**
WIND CLASSIFICATION	**N2**
EXPOSURE CLASSIFICATION CONCRETE SURFACE	**INTERNAL (A1) EXTERNAL (B1)**
THE SITE CLASSIFICATION REPORT PREPARED BY INTRAX CONSULTING ENGINEERING PTY LTD REPORT NO 248733 DATED 31/07/25. WE TAKE NO RESPONSIBILITY FOR VARIATIONS THAT MAY OCCUR DUE TO VARIATIONS IN SITE CONDITIONS.	
PROVIDE ADDITIONAL 1N12 (2000 LONG) AT ALL INTERNAL PIER LOCATION – BOTH WAYS FIXED TO UNDERSIDE OF THE TOP FABRIC MESH.	

LEGEND

- DENOTES HEBEL WALL OVER
- DENOTES TIMBER WALL OVER
- DENOTES BRICK WALL OVER
- DENOTES 3N12 OR EQUIV. (OR 3L11TM) CRACK CONTROL BARS, 2000mm LONG TIED UNDER TOP MESH.
- DENOTES 300 DEEP PODS (D1). POD SIZE: 1090x1090 (CUT PODS AS REQUIRED)
- DENOTES 225 DEEP PODS (D2). POD SIZE: 1090x1090 (CUT PODS AS REQUIRED)
- DENOTES POD STARTING POINT
- DENOTES 400 DIA. MASS CONCRETE PIERS (250 KPA MIN. BEARING CAPACITY) TO UNIFORM BEARING ON NATURAL STRATA THROUGHOUT & 400MM MIN. INTO SHALE MATERIAL. ALL PIERS: 1.5M MIN. DEEP OR NATURAL ROCK – WHICHEVER IS LESSER.

GROUND FLOOR WAFFLE SLAB LAYOUT PLAN

SCALE 1:100

1. 300mm DEEP PODS U.N.O. + SLAB THICKNESS = 90mm & PROVIDE SL82 TOP MESH (20 COVER)
2. CONCRETE STRENGTH f'c = 32 MPa (PIERS) AND f'c = 32 MPa (SLAB) MINIMUM AT 28 DAYS U.N.O.
3. LAID 0.2 MICRON HIGH IMPACT DAMP-PROOF MEMBRANE UNDER WAFFLE POD SLAB.
4. IF GROUND CONDITIONS CHANGE DURING EXCAVATION, PLEASE NOTIFY ENGINEER AND SEEK FURTHER INSTRUCTIONS.
5. IF PIER DEPTH EXCEEDS 2.5m, ENGINEER TO BE NOTIFIED FOR DECISION.
6. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS, FALLS ETC.
7. THIS DRAWINGS IS SIGNED SUBJECT TO CERTIFICATE OF INSPECTION ISSUED BY AN ENGINEER FROM THIS OFFICE. ALL PIERS AND REINFORCEMENT SHALL BE INSPECTED BY THIS OFFICE PRIOR TO PLACING CONCRETE.

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BEROWRA, NSW
FOR – VIKAS KUMAR RANGI & POONAM CHAUDHARY

BUILDER

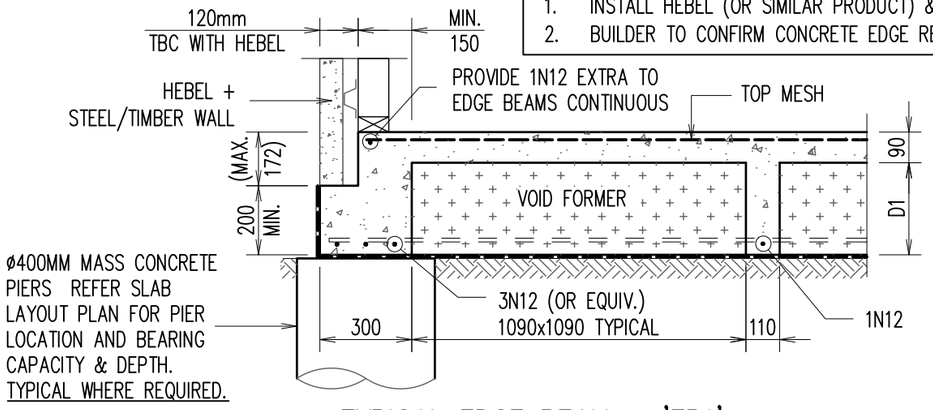
WAFFLE SLAB PLAN

SCALE: 1:100, 1:20 U.N.O.			
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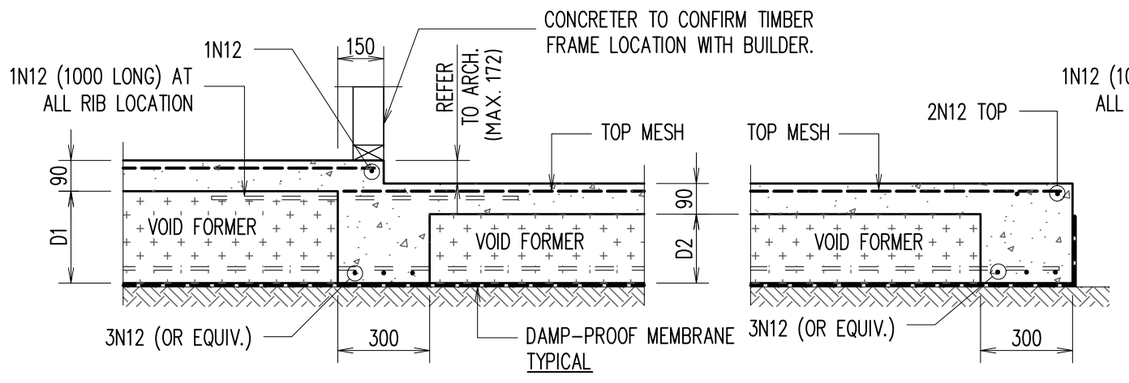
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2. BUILDER TO CONFIRM CONCRETE EDGE REBATE WITH HEBEL (OR SIMILAR PRODUCT) MANUFACTURERS DETAILS AND SPECIFICATIONS.

JOB SPECIFICATION	
POD DEPTH (D1)	300mm
POD DEPTH (D2)	225mm
SLAB THICKNESS	90mm
TOP MESH	SL82
BEAM REINFORCEMENT: 3N12 (ALTERNATIVELY 3-L11TM)	



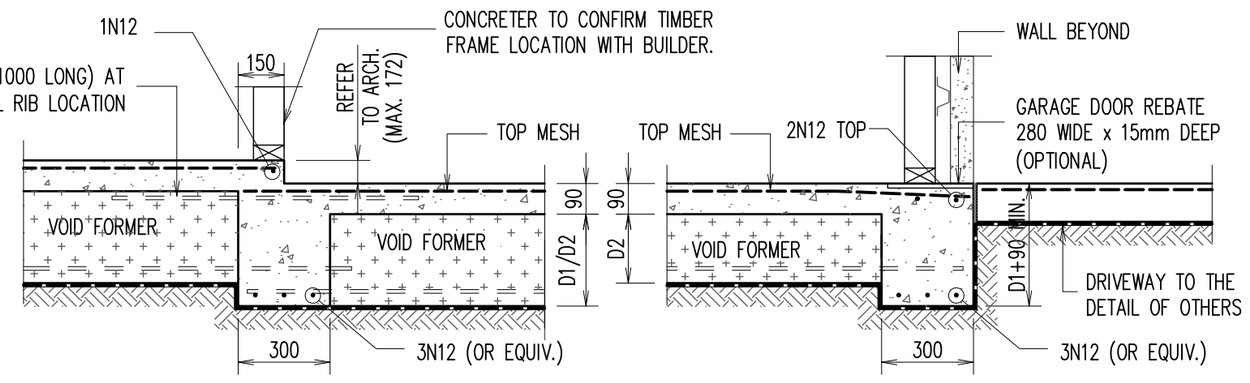
TYPICAL EDGE BEAM – 'EB1'

ALTERNATIVELY REFER TO 'DB1' FOR DETAILS



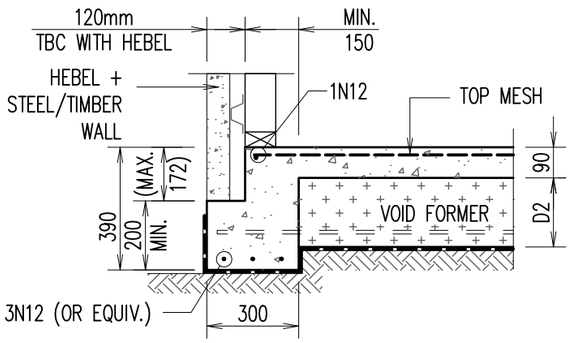
INTERNAL RIB

INTERNAL BEAM – 'IB1'



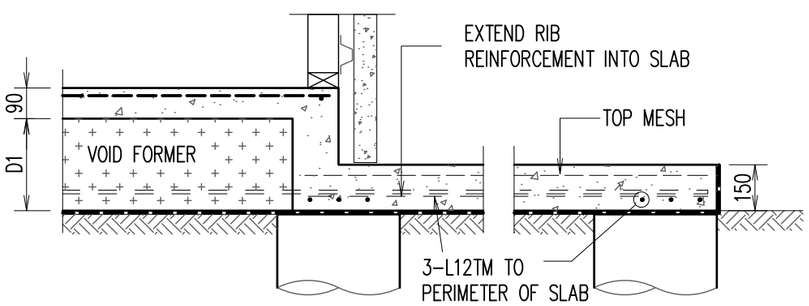
INTERNAL STEPDOWN BEAM – 'IB2'

ALTERNATIVELY REFER TO 'IB3 OR DB2' FOR DETAILS



EDGE BEAM – 'EB2'

ALTERNATIVELY REFER TO 'DB3' FOR DETAILS



INTERNAL STEPDOWN BEAM – 'IB3'

ALTERNATIVELY REFER TO 'DB2' FOR DETAILS

GARAGE EDGE BEAM – 'EB3'

GARAGE EDGE BEAM – 'EB4'

TYPICAL ATTACHED TANK/AC SLAB DETAIL

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BEROWRA, NSW
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& POONAM CHAUDHARY

BUILDER

WAFFLE SLAB DETAILS – 1

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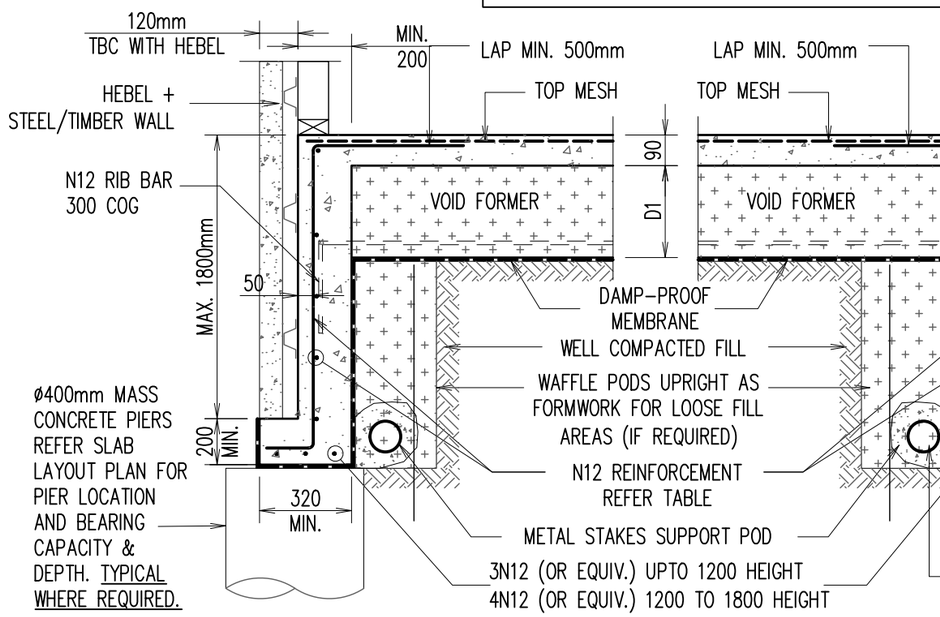
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2. BUILDER TO CONFIRM CONCRETE EDGE REBATE WITH HEBEL (OR SIMILAR PRODUCT) MANUFACTURERS DETAILS AND SPECIFICATIONS.



'EXTERNAL' DROP BEAM – 'DB1'

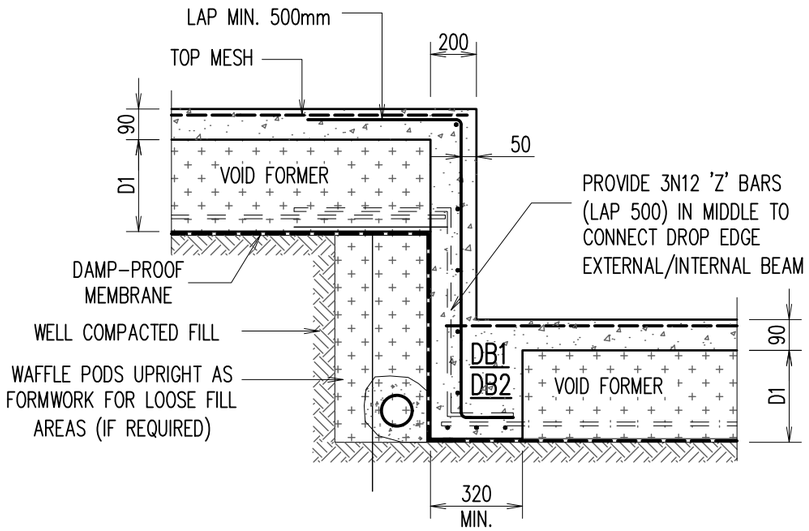
'INTERNAL' DROP BEAM – 'DB2'

'EXTERNAL' DROP BEAM – 'DB3'

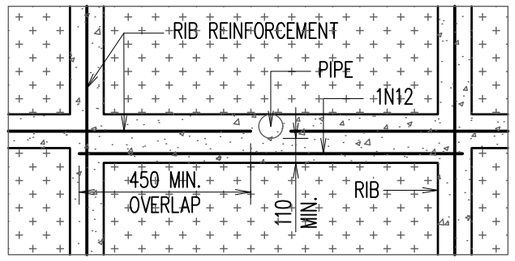
JOB SPECIFICATION	
POD DEPTH (D1)	**300mm**
POD DEPTH (D2)	**225mm**
SLAB THICKNESS	**90mm**
TOP MESH	**SL82**
BEAM REINFORCEMENT: **3N12 (ALTERNATIVELY 3-L11TM)**	

DROP EDGE BEAM REINFORCEMENT TABLE		
DROP EDGE BEAM DEPTH	HORIZONTAL REO	VERTICAL REO
UP TO 400mm	1N12	1N12
410 – 600mm	N12-400 CTS	N12-400 CTS
610 – 1200mm	N12-300 CTS	N12-300 CTS
1210 – 1600	N12-200 CTS	N12-200 CTS

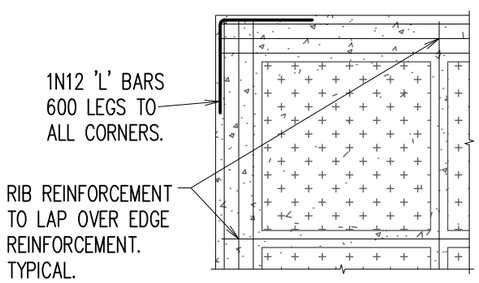
REINFORCEMENT FOR EXTERNAL BEAMS WHERE WIDTH EXCEEDS 300mm		
WIDTH	TOP STEEL	BTM STEEL EXTRA
301-330	1N12	1N12
331-440	1N12	2N12
441-550	2N12	3N12
551-660	3N12	4N12



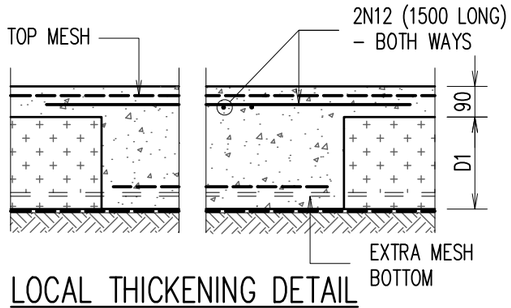
DROP EDGE BEAM TRANSITION DETAIL



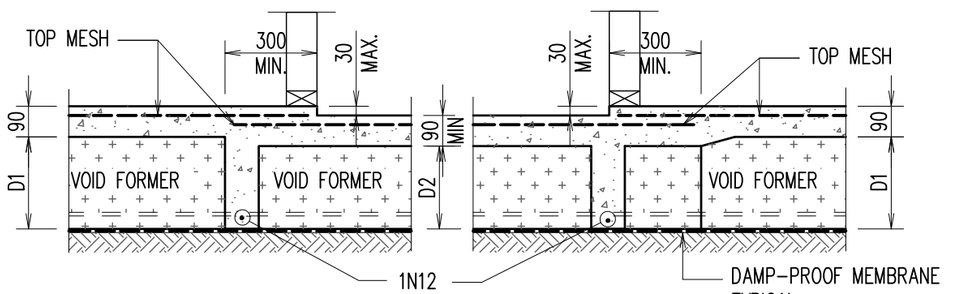
PIPE PENETRATION THROUGH RIB



TYPICAL CORNER DETAILS



LOCAL THICKENING DETAIL



TYPICAL WET AREA STEP DETAIL

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BUILDER

WAFFLE SLAB DETAILS – 2

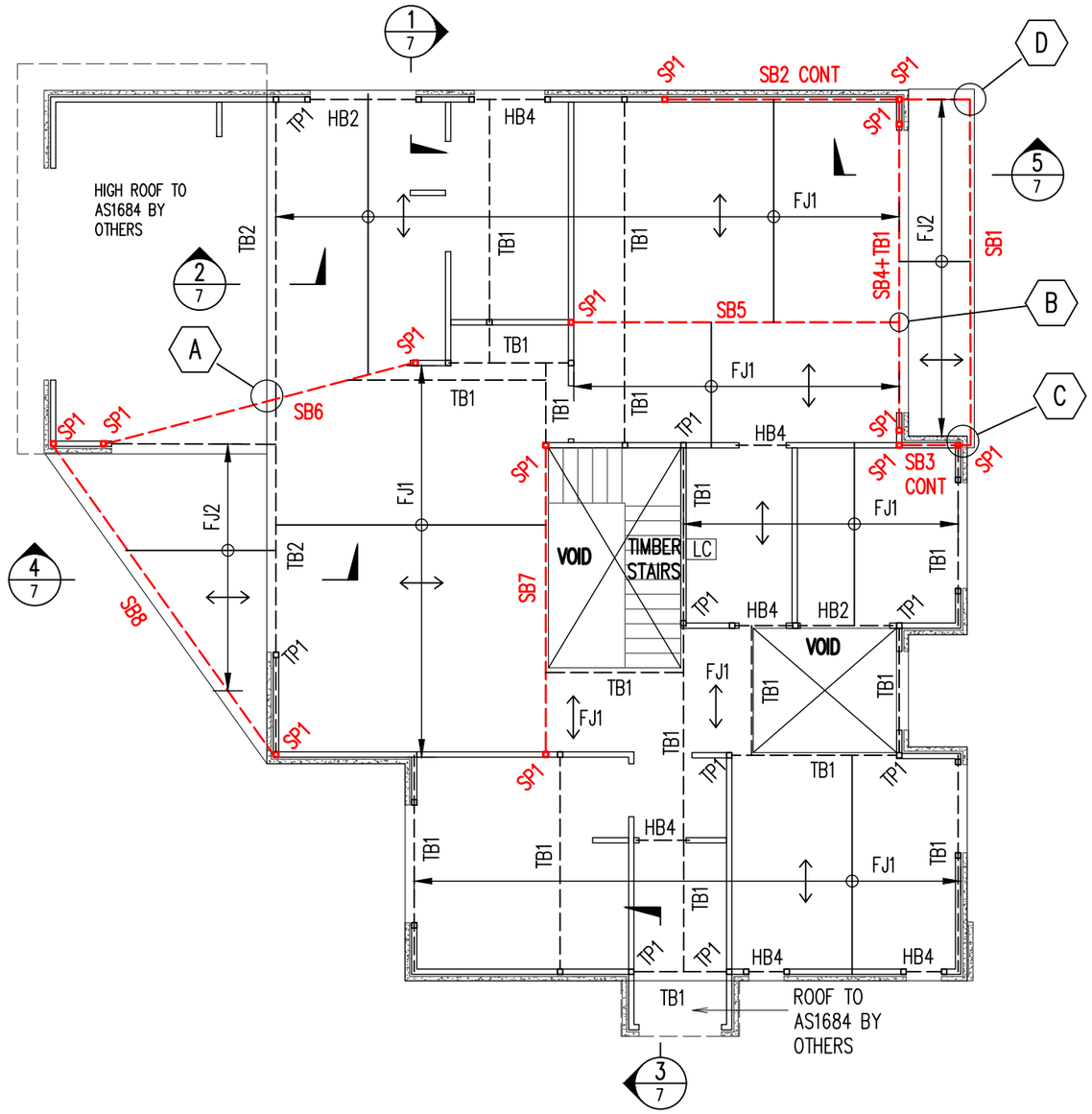
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FIRST FLOOR – STEEL BEAM AND FLOOR JOIST LAYOUT PLAN

SCALE 1:100

1. DRAWINGS TO BE READ IN CONJUNCTION WITH ARCHITECTURAL.
2. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS, FALLS ETC.
3. PROVIDE DOUBLE FLOOR JOISTS UNDER ALL WALLS RUNNING PARALLEL TO JOISTS
4. TB1 TO BE PROVIDED ON OVER ALL OPENINGS
5. PROVIDE DOUBLE STUD UNDER ALL TIMBER AND HEADER BEAM.

MEMBER SCHEDULE		
MARK	SIZE	COMMENTS
SB1,SB2	300 PFC	300 GRADE
SB3,SB4,SB8	300 PFC	300 GRADE
SB5,SB6,SB7	310UB46.2	300 GRADE
SP1	89 x 89 x 5.0 SHS	350 GRADE
FJ1	HJ 300 63 HYJOIST	AT 450 CTS OR MANUFACTURES DETAILS
FJ2	190 X 45 MGP10 OR F7 LVL	AT 450 CTS
TB1,HB1	300 x 63 HYSPAN	TIMBER BEAM
TB2	2/300 X 63 HYSPAN	M10 4.6/S @ 450 CTS TOP & BOTTOM
HB2	2/190 X 45 MGP10 OR F7	TIMBER HEADER BEAM
HB3,TB3	240 X 63 HYSPAN	TIMBER BEAM
HB4	190 X 45 MGP10 OR F7	TIMBER HEADER BEAM
TB4	360 X 63 HYSPAN	TIMBER BEAM
TB,HB	STRUCTURAL TIMBER BEAM	TO MANUFACTURES DETAILS
TP1	STRUCTURAL TIMBER POST	TO MANUFACTURES DETAILS

NOTE:

1. IF IN DOUBT FOR LINTEL SIZES OVER REMAINING WINDOW/DOOR OPENINGS CONTACT STRUCTURAL ENGINEER.
2. FIRST FLOOR PARAPET WALLS AND SUPPORT DETAILS BY OTHERS.
3. FLOOR JOIST ARE NOT DESIGN TO TAKE ROOF LOAD, FOR CONVENTIONAL ROOF, CARPENTER NEED TO PROVIDE TIMBER BEAM AS PER AS1684.
4. TIMBER TRUSS ROOF AND WALLS BRACING AND TIE DOWN DETAILS BY OTHERS & COMPLY WITH RELEVANT SECTIONS OF AS1684.
5. INSTALL HEBEL (OR SIMILAR PRODUCT) & STUD WALLS AS PER MANUFACTURERS DETAILS AND SPECIFICATIONS.
6. PROVIDE ARTICULATION JOINTS – LOCATION & DETAIL AS PER HEBEL MANUFACTURE DETAILS.

LEGEND

- DENOTES BRICK WALL UNDER
- DENOTES HEBEL WALL OVER
- DENOTES TIMBER WALL UNDER
- DENOTES FLOOR JOIST DIRECTION

CARPENTER TO PROVIDE ADEQUATE TIMBER BEAM UNDER ALL FIRST FLOOR WALLS RUNNING PARALLEL TO FLOOR JOIST & ROOF POINT LOAD AS PER AS1684 FOR CUT ON SITE ROOF.

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FIRST FLOOR STEEL/JOIST PLAN

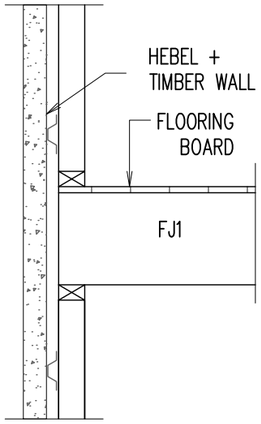
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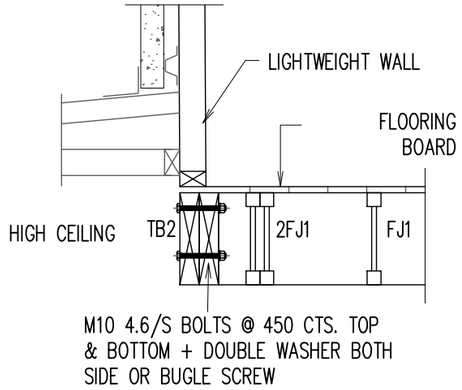
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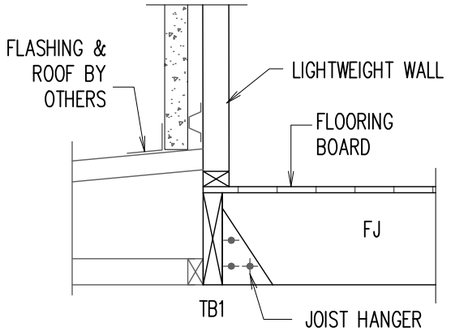
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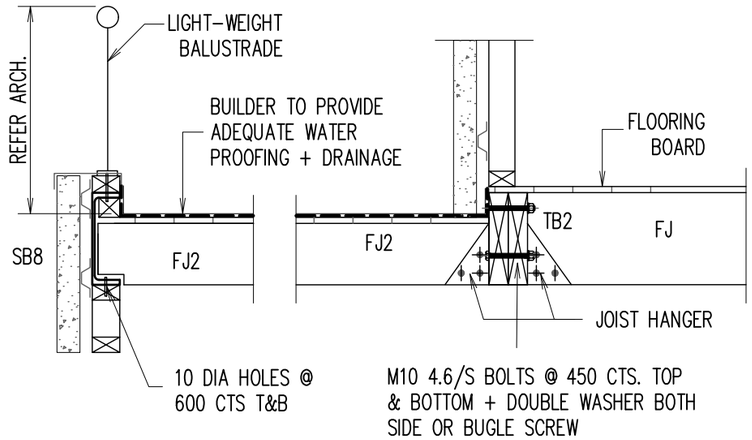
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SCALE 1:20



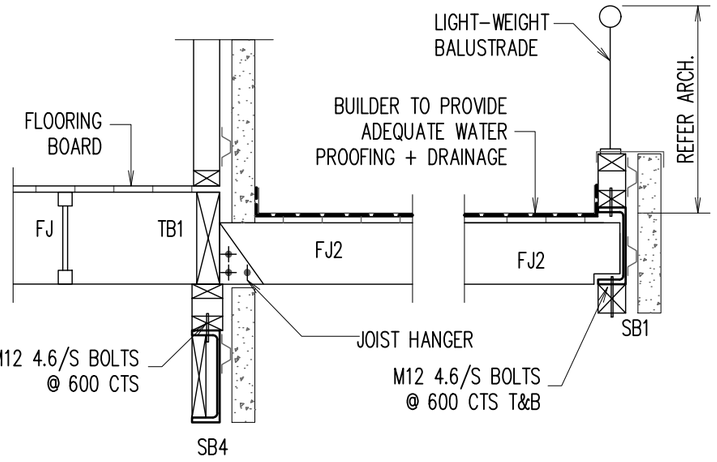
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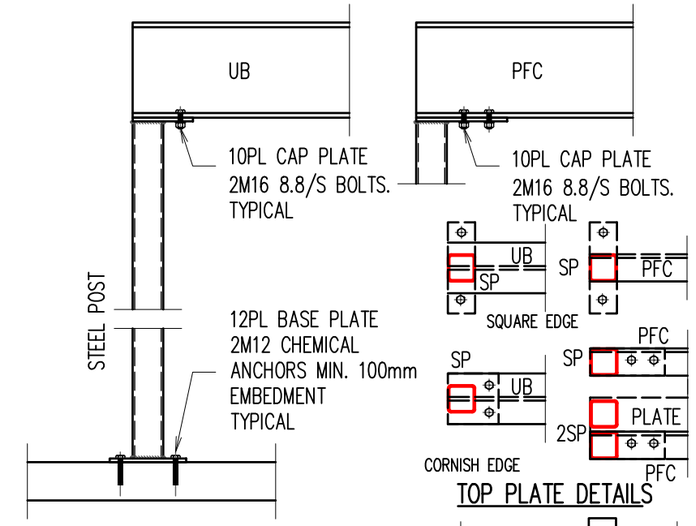
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SECTION 4/6
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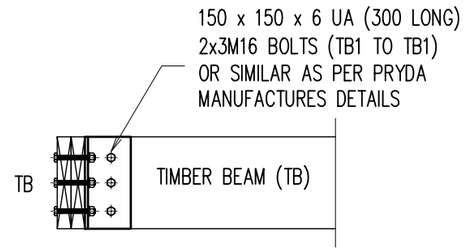


SECTION 5/6
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TYPICAL POST TO SLAB DETAIL

NOTE: POSTS TO BE SECURELY BRACED TO STUD FRAMEWORK



TYPICAL TIMBER BEAM
(TB) TO TB DETAIL

A3 0 1 2 3 4 5 6 7 8 9 10

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PROPOSED RESIDENCE
LOT 3 (NO. 75) BEROWRA WATERS ROAD
BEROWRA, NSW
FOR – VIKAS KUMAR RANGI
& POONAM CHAUDHARY

BUILDER

STEEL WORK DETAIL

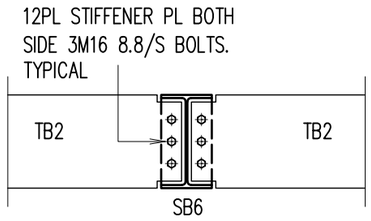
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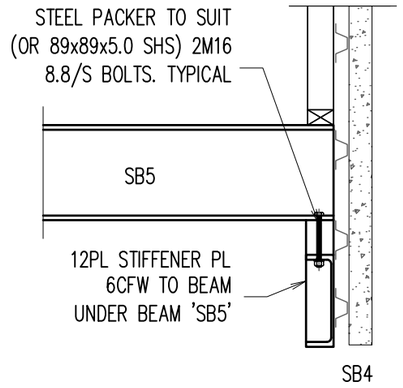
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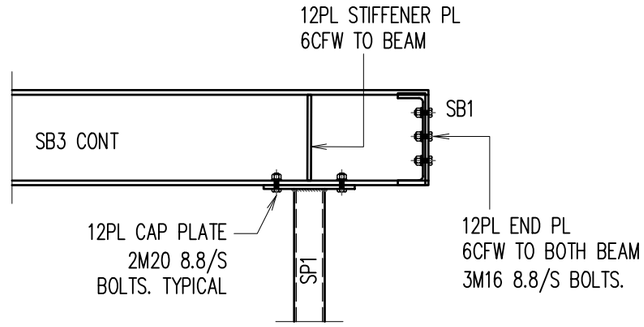
DETAIL – A

SCALE 1:20



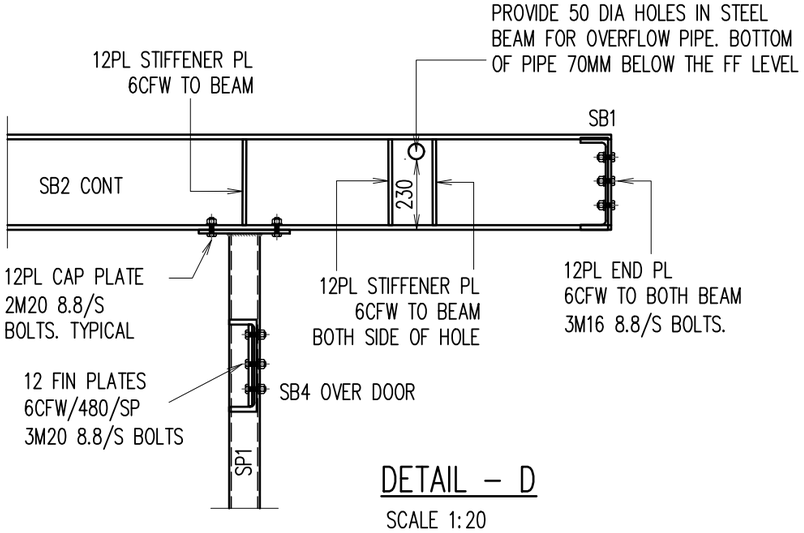
DETAIL – B

SCALE 1:20



DETAIL – C

SCALE 1:20



DETAIL – D

SCALE 1:20

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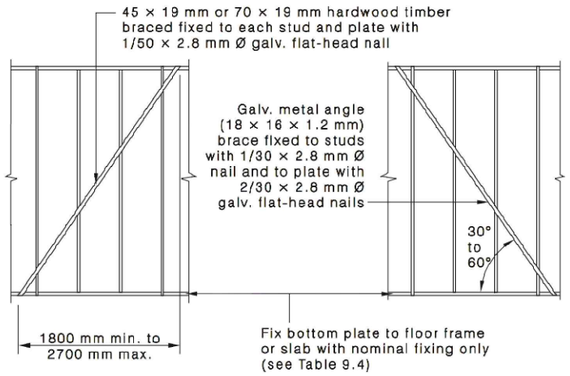
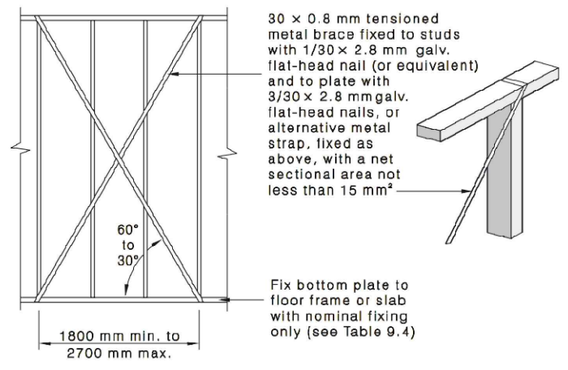
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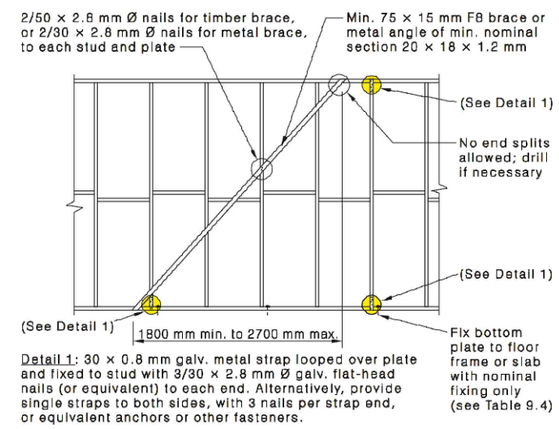
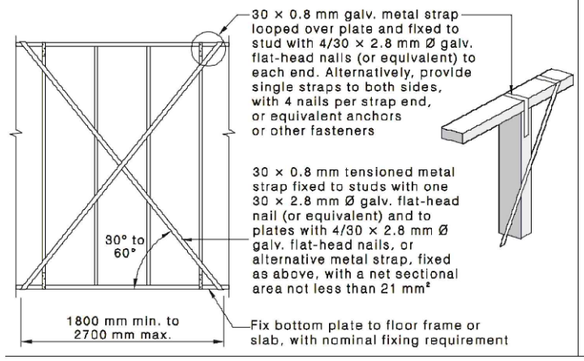
BUILDER

STEEL WORK DETAIL

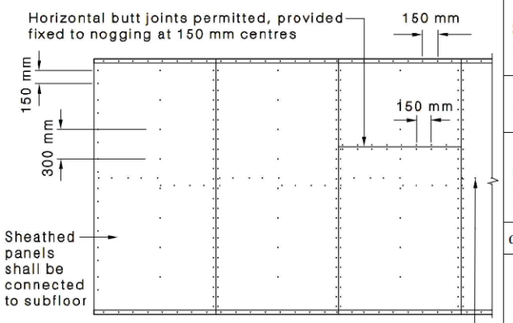
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Type of bracing	Bracing capacity kN/m
(a) *Two diagonally opposed timber or metal angle braces*	
	0.8
NOTE: All flat-head nails shall be galvanized or equivalent.	
(b) *Metal straps—Tensioned*	
	1.5

Type of bracing	Bracing capacity kN/m
(c) *Timber and metal angle braces* The maximum depth of a notch or saw-cut shall not exceed 20 mm. Saw-cuts studs shall be designed as notched.	
	1.5
(d) *Metal straps—Tensioned—With stud straps*	
	3.0

(g) *Plywood* Plywood shall be nailed to frame using 30 mm × 2.8 mm Ø galvanized flat-head nails or equivalent.



Stress grade	Minimum plywood thickness, mm: Stud spacing mm 450	600
No nogging (except horizontal butt joints)		
F8	7	9
F11	4.5	7
F14	4	6
F27	3	4.5
One row of nogging		
F8	7	7
F11	4.5	4.5
F14	4	4
F27	3	3

3.4

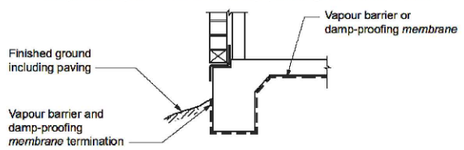
NOTES:
1 For plywood fixed to both sides of the wall, see Clauses 8.3.6.5 and 8.3.6.10.
2 No other rods or straps are required between top or bottom plate.
3 Fix bottom plate to floor frame or slab with nominal fixing only (see Table 9.4).

TYPICAL WALL BRACING DETAIL

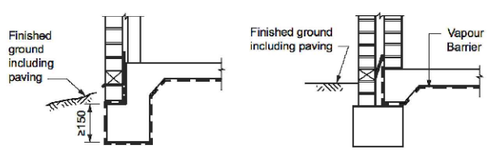
NOTE: IN ACCORDANCE WITH REQUIREMENTS OF AS1684 NATIONAL FRAMING CODE.

NOTE :
ALL BRACING TO COMPLY WITH RELEVANT SECTIONS OF AS1684 RESIDENTIAL TIMBER FRAMED CONSTRUCTION, FOR FURTHER INFORMATION REFER TO CLAUSE 8.3.6 OF PART 2 AS1684.2–2010

Figure 4.2.8: Acceptable vapour barrier and damp-proofing membrane location

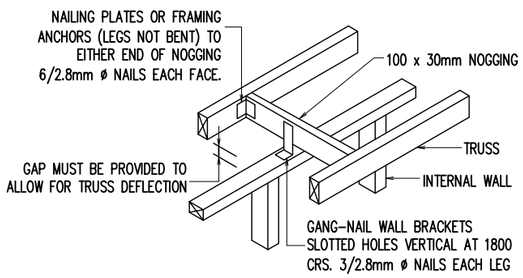


(a) Minimum rebate for cavity masonry or veneer wall

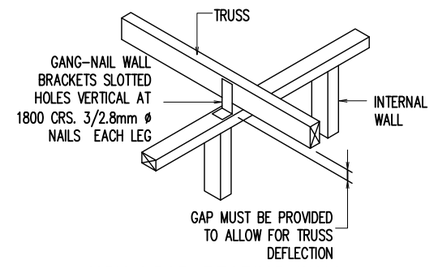


(b) Deep edge rebate alternative

(c) Masonry alternative

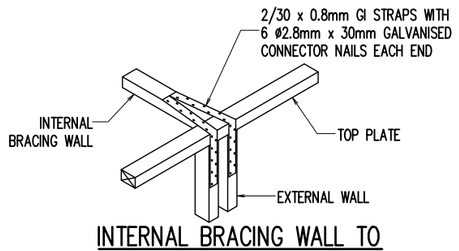


NON LOAD BEARING WALL PARALLEL TO TRUSSES



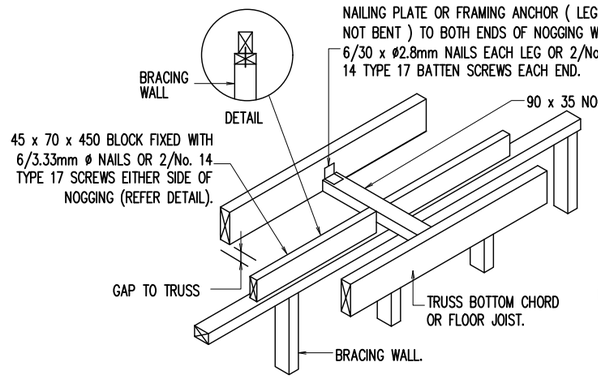
NON LOAD BEARING WALL PERPENDICULAR TO TRUSSES

SCALE NTS

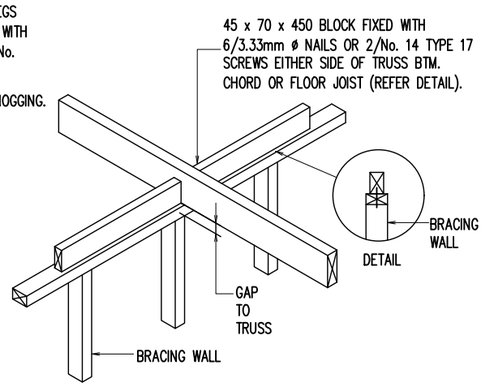


INTERNAL BRACING WALL TO EXTERNAL WALL CONNECTION

SCALE NTS



PARALLEL TO TRUSS



PERPENDICULAR TO TRUSS

TIMBER BRACING WALL TOP CONNECTION DETAIL

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TYPICAL BRACING DETAIL

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